

Pharmaceutical Reform and Pharmacy Entry

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County-level Empirical Setting

- ▶ Unit of analysis: county-by-year panel, 2012–2022.
- ▶ Main controlled sample: 2012–2019, with population, nighttime lights, and road-density controls.
- ▶ Main treatment indicator:

$$Reform_{ct} = \mathbf{1}\{t \geq PolicyYear_c\}.$$

- ▶ Baseline FE specification:

$$Y_{ct} = \beta_0 + \beta_1 Reform_{ct} + \beta X_{ct} + \alpha_c + \lambda_t + \varepsilon_{ct}.$$

- ▶ Controls include log population, log nighttime lights, and log road density where available.
- ▶ Standard errors are clustered at the county level.

County Sample and Descriptive Statistics

Variable	N	Mean	SD	P25	Median	P75
Pharmacy count	17,004	95.22	108.77	26	63	128
Log pharmacy count	15,512	4.14	1.15	3.54	4.26	4.92
Annual change in count	14,279	23.15	40.31	1	12	33
Annual change in log count	12,780	0.22	0.53	-0.04	0.16	0.43
Has pharmacy	17,004	0.91	0.28	1	1	1
Chain pharmacy count	17,004	8.09	27.30	0	0	1
Non-chain pharmacy count	17,004	87.13	96.55	25	59	117
DID	17,004	0.64	0.48	0	1	1

County TWFE DID: Pharmacy Stock and Annual Change

	Count	Count + controls	Annual Δ count
Post policy	7.652*** (1.474)	5.449*** (1.454)	19.892*** (2.124)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Controls	No	Yes	Yes
N	16,961	16,961	14,279
R ²	0.824	0.845	0.334

TWFE DID: Log Count and Extensive Margin

	ln(count)	Has pharmacy
Post policy	0.020 (0.014)	0.030*** (0.004)
County FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
N	15,469	16,961
R ²	0.882	0.664

Continuous DID

	Count	ln(count)
Post policy $\times$ ln(public hospitals in 2012 + 1)	20.672*** (3.776)	-0.068** (0.024)
County FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
N	16,961	15,469
R ²	0.847	0.882

CSDID and PPML

	Pharmacy count	ln(count)
ATT	-2.178 (2.077)	0.010 (0.032)
Pre-average	-2.403 (2.696)	0.038 (0.077)
Post-average	10.002* (4.610)	0.084 (0.084)
Controls	Yes	Yes
Control group	Not-yet-treated	Not-yet-treated

PPML count FE	
Post policy	-0.014 (0.012)
County FE	Yes
Year FE	Yes
Controls	Yes
N	16,947

County Distance to Nearest Hospital

	Mean km	Share ≤ 2 km	≤ 2 km count	> 2 km count
Post policy	0.000 (0.124)	0.001 (0.005)	5.314*** (1.021)	0.135 (0.785)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	15,469	15,469	16,961	16,961
R ²	0.666	0.740	0.849	0.763

Distance measures are based on each pharmacy's nearest hospital of any ownership type.

County Pharmacies Near Public and Private Hospitals

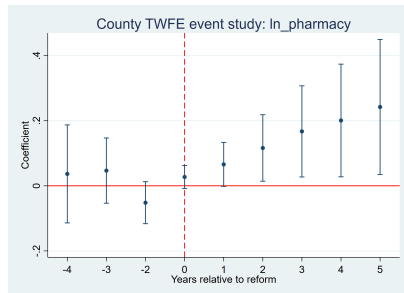
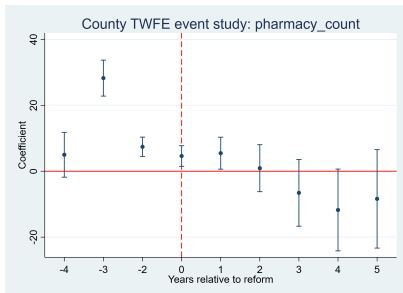
	Public $\leq 2\text{km}$	Private $\leq 2\text{km}$	Public share	Private share
Post policy	4.812*** (1.063)	9.529*** (1.077)	-0.020** (0.007)	0.011* (0.005)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	15,469	15,469	15,469	15,469
R ²	0.783	0.870	0.703	0.847

Public/private 2km counts are not mutually exclusive: a pharmacy can be within 2km of both a public and a private hospital.

County Chain vs Non-chain Outcomes

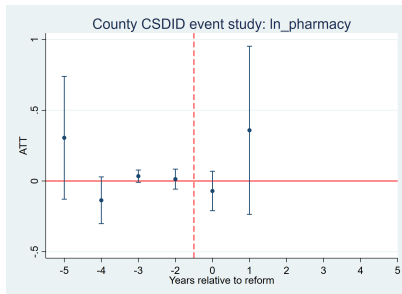
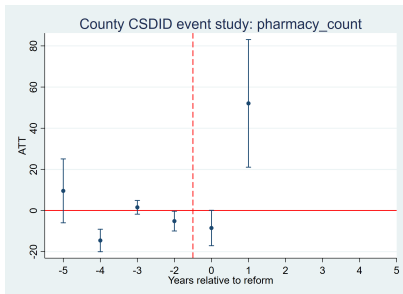
	Chain count	Non-chain count	Chain share
Post policy	2.712*** (0.359)	2.737* (1.280)	0.009*** (0.002)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
N	16,961	16,961	16,961
R ²	0.642	0.826	0.663

County TWFE Event-study Diagnostics



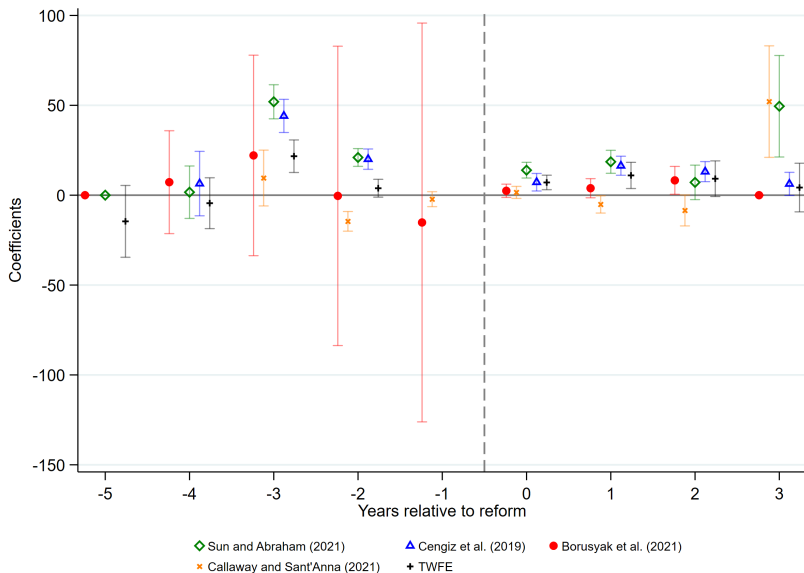
Standard TWFE event-study graphs with county and year fixed effects. Relative year -1 is omitted.

County CSDID Event-study Diagnostics

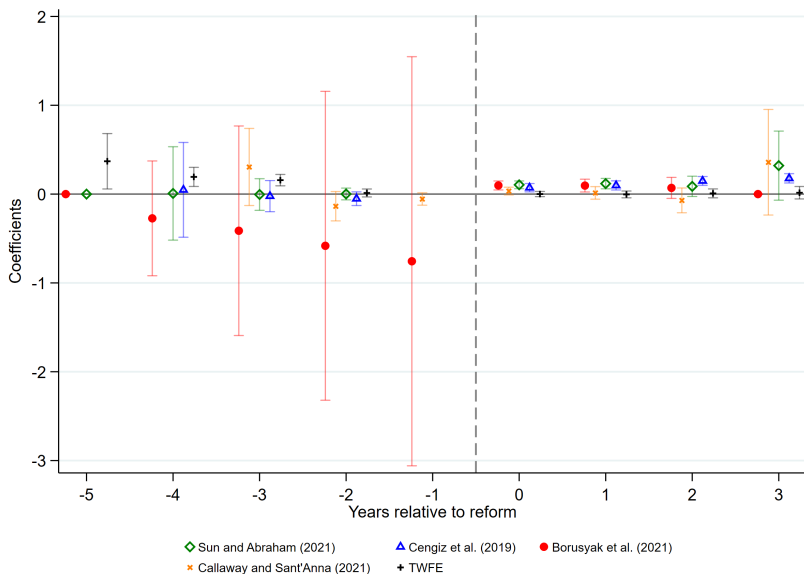


CSDID uses not-yet-treated counties as controls. The updated event window is $[-5, +5]$.

County Five-estimator Event-study Robustness: Count



County Five-estimator Event-study Robustness: $\ln(\text{count})$



Hospital-level DDD Specification

- ▶ Unit of analysis: hospital-by-year panel with pharmacy outcomes measured within 1km, 2km, and 3km buffers.
- ▶ Treatment contrasts public hospitals with non-public hospitals after local reform implementation:

$$Y_{ict} = \beta_0 + \beta(\text{Public}_i \times \text{Post}_{ct}) + \theta \text{Post}_{ct} + X'_{ict}\gamma + \alpha_i + \lambda_t + \varepsilon_{ict}.$$

- ▶ β is the differential post-reform change around public hospitals.
- ▶ $\theta + \beta$ is the post-reform change around public hospitals relative to the pre-period.
- ▶ Controls include nighttime lights around hospitals where available. Standard errors are clustered at the county level.

Hospital DDD: Pharmacy Count

	1km	2km	3km
Public $\times$ Post	-0.403* (0.192)	-1.272** (0.491)	-2.148* (0.856)
Post policy	1.081*** (0.148)	4.171*** (0.430)	8.757*** (0.814)
Hospital FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
NTL control	Yes	Yes	Yes
N	164,674	164,674	164,674
R ²	0.895	0.906	0.911

The public-hospital interaction is negative, while the post-policy level shift is positive.

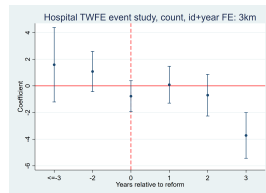
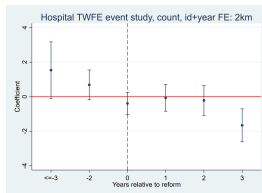
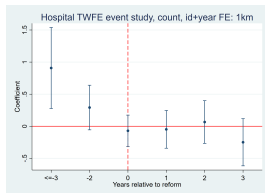
Hospital Log and PPML Outcomes

	ln(count)			PPML count		
	1km	2km	3km	1km	2km	3km
Public $\times$ Post	-0.005 (0.017)	0.025 (0.017)	0.018 (0.017)	-0.036* (0.015)	-0.029* (0.013)	-0.027* (0.012)
Post policy	0.012 (0.015)	0.013 (0.015)	0.007 (0.016)	– –	– –	– –
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
NTL control	Yes	Yes	Yes	Yes	Yes	Yes
N	134,788	147,008	151,443	147,711	156,544	159,753
R ²	0.913	0.938	0.944	–	–	–

Hospital CSDID Average ATT

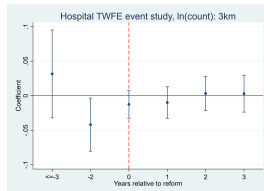
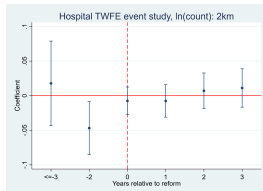
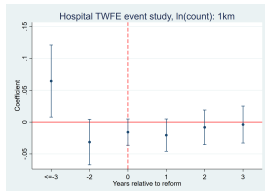
	1km	2km	3km
Count	-0.004 (0.169)	-0.447 (0.433)	-1.127 (0.755)
ln(count)	-0.009 (0.017)	0.004 (0.016)	-0.006 (0.015)

Hospital TWFE Event Study: Count, FE



Window is restricted to event time $[-3, 3]$; relative year -1 is omitted. Models absorb hospital and year fixed effects.

Hospital TWFE Event Study: $\ln(\text{count})$, FE



$\ln(\text{count}+1)$ event-study graphs are omitted. Models absorb hospital and year fixed effects.

How the County and Hospital Results Connect

- ▶ **County-level estimates show market expansion.** Pharmacy counts rise after reform in the county FE models, and annual count changes increase sharply.
- ▶ **Distance results do not imply broad movement away from hospitals.** Average distance and the within-2km share are almost unchanged, while counts within 2km rise.
- ▶ **Public-hospital proximity rises in absolute terms but weakens in relative terms.** Pharmacies within 2km of public hospitals increase, but the public-hospital share falls while the private-hospital share rises.
- ▶ **Hospital DDD is a relative comparison.** The negative Public $\times$ Post coefficient means public hospitals experience smaller growth than non-public hospitals, not necessarily an absolute decline around public hospitals.
- ▶ **Interpretation.** Reform appears to expand the county pharmacy market overall, with growth near hospitals, but new entry is relatively less concentrated around public hospitals than around private or non-public hospital areas.